2 Definitions

L BP is the length between perpendiculars

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SOURCE: MSC.141(76).pdf
prove capable of withstanding the seaway define d in paragraph 4 below, in the worst damage case scenario.

2 Definitions

L BP is the length between perpendiculars
H S is the significant wave height
B is the moulded breadth of the ship
T P is the peak period
T Z is the zero crossing period

3 Ship model

3.1.1 The model should copy the actual ship for both outer configuration and internal arrangement – in particular all damaged spaces having an effect on the process of flooding and shipping of water. Intact dr aught, trim, heel and limiting operational KG corresponding to the worst damage case should be used. Furthermore, the test case(s) to be considered should represent the worst damage case(s) defined in accordance with SOLAS regulation II-1/8.2.3.2 (SOLAS 90) with regard to the total area under the positive GZ curve and the centreline of the da mage opening should be located within the following range:

.1 ± 35 % LBP from midship;

.2 an additional test will be requi red for the worst damage within ±10% L BP from midship if the damage case referred to in .1 is outside of ±10% LBP from midship.

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.1 ± 35 % LBP from midship;

.2 an additional test will be requi red for the worst damage within ±10% L BP from midship if the damage case referred to in .1 is outside of ±10% LBP from midship.

3.2 The model should comply with the following:

.1 the length between perpendiculars (L BP) should be at least 3 m or a length corresponding to a model scale of 1:40, wh ichever is greater, and the vertical extent up to at least 3 superstructure standard heights above the bulkhead (freeboard) deck;

.2 hull thickness of flooded spaces should not exceed 4 mm;

.3 in both intact and damaged conditions, the model should satisfy the correct displacement and draught marks (TA, TM, TF, port and starboard) with a maximum tolerance in any one draught mark of +2 mm. Draught marks forward and aft should be located as near FP and AP as practicable;

.4 all damaged compartments and ro-ro sp aces should be modelled with the correct

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tolerance in any one draught mark of +2 mm. Draught marks forward and aft should be located as near FP and AP as practicable;

.4 all damaged compartments and ro-ro sp aces should be modelled with the correct surface and volume permeabilities (actual values and distributions) ensuring that floodwater mass and mass distribution are correctly represented;

.5 the actual ship characteristics of mo tion should be modelled properly, paying particular attention to the intact GM to lerance and radii of gyration in roll and pitch motion. Both radii should be measured in air and be in the range of 0.35B to 0.4B for roll motion, and 0.2LOA to 0.25LOA for pitch motion;

.6 main design features such as waterti ght bulkheads, air escapes, etc., above and below the bulkhead deck that can result in asymmetric flooding should be modelled properly as far as practicable, to represent the real situation. Ventilating and cross-flooding arrangements should be constructed to a minimum cross section of 500 mm2 ;

.7 the shape of the damage opening at side should be as follows:

.1 trapezoidal profile with side at 15° slope to the vertical and the width at

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and cross-flooding arrangements should be constructed to a minimum cross section of 500 mm2 ;

.7 the shape of the damage opening at side should be as follows:

.1 trapezoidal profile with side at 15° slope to the vertical and the width at the design waterline defined according to SOLAS regulation II-1/8.4.1;

.2 isosceles triangular profile in the horizontal plane with the height equal to B/5 according to SOLAS regulation II-1 /8.4.2. If side casings are fitted within B/5, the damaged length in wa y of the side casings should not be

less than 25 mm;

.3 notwithstanding the provisions of subparagraphs .7.1 and .7.2 above, all compartments taken as damaged in calculating the worst damage case(s) referred to in paragraph 3.1 should be flooded in the model tests.

3.3 The model in the flooded equilibrium cond ition should be heeled by an additional angle corresponding to that induced by the heeling moment M h = max(Mpass; Mlaunch)-Mwind, but in no case should the final heel be less than 1° towards damage. Mpass, Mlaunch and Mwind are as specified in SOLAS regulation II-1/8.2.3.4. For existing ships this angle may be taken as 1.

4 Procedure for experiment

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SOURCE: MSC.141(76).pdf
case should the final heel be less than 1° towards damage. Mpass, Mlaunch and Mwind are as specified in SOLAS regulation II-1/8.2.3.4. For existing ships this angle may be taken as 1.

4 Procedure for experiment

4.1 The model should be tested in a long-cres ted irregular seaway defined by the JONSWAP spectrum with significant wave height H S, a peak enhancement factor =3.3 and a peak period TP = 4 , (TZ = TP /1.285). HS is the significant wave height for the area of operation, which is not exceeded by a probability of more than 10% on a yearly basis, but limited to a maximum of 4 m.

Furthermore,

.1 the basin width should be sufficient to a void contact or other interaction with the sides of the basin and is recommended not to be less than LBP + 2 m;

.2 the basin depth should be sufficient fo r proper wave modelling but should not be less than 1 m;

.3 for a representative wave realisation to be used, measurements should be

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.2 the basin depth should be sufficient fo r proper wave modelling but should not be

less than 1 m;

.3 for a representative wave realisation to be used, measurements should be performed prior to the test at 3 different locations within the drift range;

.4 the wave probe closer to the wave ma ker should be located at the position where the model is placed when the test starts;

.5 variation in H S and TP should be within ±5% for the three locations; and

.6 during the tests, for approval pur poses, a tolerance of +2.5% in H S, ±2.5% in T P and ±5% in T Z should be allowed with reference to the probe closer to the wave maker.

4.2 The model should be free to drift and placed in beam seas (90° heading) with the damage facing the oncoming waves, with no mooring system permanently attached to the model used. To maintain a beam sea heading of approximate ly 90° during the model test the following requirements should be satisfied:

.1 heading control lines, intended for minor adjustment, should be located at the centre line of the stem and stern, in a sy mmetrical fashion and at a level between

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